Homework 3

Bonds and Risk

Covers: Lectures 6: Risk and risk aversion and 7:an variance tradeoffs

1 Financial Literacy

Answer these questions in 3 steps:

1. Give an answer to the question.
2. Look over the material assigned for class and find a definition.
3. Modify, if necessary, your first answer.

The goal is not for you to memorize a given answer but to ensure you can explain the concept to someone. If you can't, then you do not fully understand the concept.

a. **Coupon:** The interest rate paid on a bond.

b. **Bid-ask spread:** The difference between the highest price a buyer is willing to pay for a security and the lowest price a seller is willing to accept

c. **States of the world:** The different possible values of an asset. Each possible value represents a possible state of the world.

d. **Value additivity:** The value of the bond has to be the sum of the value of cash flows.

e. **Broker:** A person or company that buys and sells securities for clients

f. **Bank assets and liabilities:** Loans

g. **Bank liabilities:** Customer deposits and bonds.

h. **Risk neutrality:** The value of a security is its expected value.

2.2 Thought problems

1. Suppose you are offered a set of zero-coupon bonds that all mature in two years. They have a probability (p) of going into default and loose all their value. Give a formula that shows how the default probability interacts with the discount rate. Then interpret your formula to tell us about the relationship between yields and risk.

   Setup. Consider a zero-coupon bond that matures in two years and pays face value F if it does not default. With probability p, the bond defaults and pays 0; with probability $1-p$, it repays in full. Let r denote the annual risk-free discount rate.

   Pricing with default risk. The expected payoff at maturity is $(1-p)F$. Discounting this payoff back to the present yields the bond price

   $$P = \frac{(1-p)F}{(1+r)^2}. \tag{1}$$

   Implied yield. Let y denote the bond's yield to maturity, defined by the standard zero-coupon pricing formula

   $$P = \frac{F}{(1+y)^2}. \tag{2}$$

   Equating the two expressions for P gives

   $$\frac{F}{(1+y)^2} = \frac{(1-p)F}{(1+r)^2}. \tag{3}$$

   Solving for y yields

   $$(1+y)^2 = \frac{(1+r)^2}{1-p}, \tag{4}$$

   and hence

   $$y = \frac{1+r}{\sqrt{1-p}} - 1. \tag{5}$$

   Interpretation. For a fixed risk-free rate r, a higher default probability p lowers the bond price and increases the yield y. The difference between y and r represents compensation for default risk: as default risk rises, investors require a higher promised yield to hold the bond.

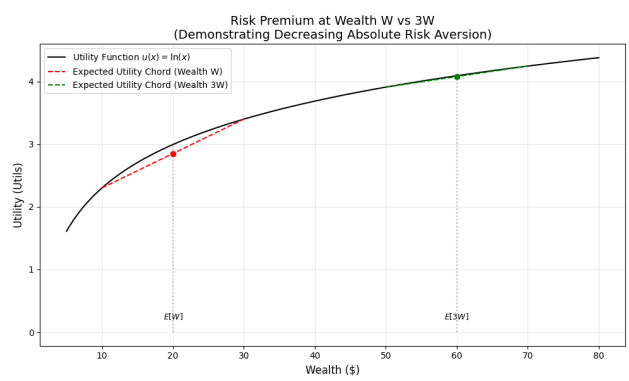
2. You are at dinner and are discussing two \$10,000 investments. One has high variance and a high expected turn, the other has low variance and low return. Some of the

guests say they do not care about variance, just return. What do you deduce about them? Other guests strongly prefer the low-V-low-R asset. What do you deduce about them? Now suppose they are only $100 investments might some of your guests change their mind?

High-variance, high-return vs. low-variance, low-return. If some guests claim that they only care about expected return and do not care about variance, we deduce that they behave as *risk-neutral* investors over stakes of $10,000. Their preferences are consistent with maximizing expected value rather than expected utility with concavity. By contrast, guests who strongly prefer the low-variance, low-return investment reveal *risk aversion*. They are willing to sacrifice expected return in order to reduce uncertainty, consistent with diminishing marginal utility of wealth. A very strong preference for the low-variance asset may also reflect heightened sensitivity to downside risk or loss aversion.

Effect of smaller stakes. If the investments are only $100, some guests may change their preferences. For small stakes, even risk-averse individuals may behave approximately risk-neutrally because utility is locally close to linear. As a result, they may become more willing to choose the high-variance, high-return investment. Truly risk-neutral individuals, however, would not change their ranking solely due to the scale of the investment.

3. Give a graphical explanation as to why an individual with wealth W will pay less than the expected value to hold a given risk asset. Then show that an individual with wealth 3W will pay more for the risky asset than the individual with wealth W.



- Concavity explains why people pay less than expected value.

- Decreasing absolute risk aversion explains why richer individuals pay more for the same risk.

4. Suppose a corporation has \$10 million of debt and 10 million dollars of equity the company is going to liquidate in 2 years. (c1) Give an explanation as to why its bonds are less risky than its debt. (c2) Is this true for all the bonds of all companies relative to their own stock? (c3) Is it also true across companies?

(c1) Within the firm. The firm has \$10 million in debt and \$10 million in equity and will be liquidated in two years. Bondholders are senior claimants and are paid first from the liquidation value, while equity holders are residual claimants. As a result, bond payoffs are more stable and equity absorbs more downside risk. Hence, the firm's bonds are less risky than its equity.

(c2) Bonds vs. equity of the same firm. In general, a firm's bonds are less risky than its stock because debt has priority in liquidation. However, this need not hold for subordinated or highly risky bonds, or when the firm is close to default.

(c3) Across firms. The ranking does not hold across companies. A bond issued by a risky firm can be more risky than the stock of a safer firm. Risk depends on firm fundamentals and leverage, not on whether the claim is debt or equity.

5. Insurance. Why might earthquake insurance be more expensive than life insurance? Be sure to define cost in a way that makes it comparable across risks. If the market for insurance is competitive, why might insurance rates rise after events (fires, accidents...) what makes life insurance different in this respect from life insurance.

Why earthquake insurance may be more expensive than life insurance. Define the *cost* of insurance as the *premium per dollar of expected payout*, i.e.,

$$\text{Loading} = \frac{\text{Premium} - \mathbb{E}[\text{Payout}]}{\mathbb{E}[\text{Payout}]}.$$

Earthquake insurance tends to have a higher loading than life insurance because earthquake losses are *highly correlated* across policyholders and occur in rare but catastrophic events. This makes risks difficult to diversify and requires insurers to hold costly capital or reinsurance. Life insurance risks, by contrast, are largely idiosyncratic

and predictable in aggregate, allowing insurers to pool risk efficiently and operate with lower loadings.

Why insurance rates may rise after events in a competitive market. Even in a competitive market, insurance premiums may rise after fires, accidents, or natural disasters because such events update insurers' beliefs about true risk (Bayesian learning) or reduce insurer capital, increasing the marginal cost of bearing risk. Higher expected losses or higher capital costs imply higher break-even premiums.

Why life insurance is different. Life insurance is based on stable mortality tables and largely independent risks across individuals. A single death event provides little new information about population-wide mortality risk and does not significantly affect insurers' aggregate exposure. As a result, life insurance premiums are relatively stable over time and do not jump after isolated events.

2 Bank Balance Sheet

The Pasatech Bank makes three kinds of loans, 6 months, 3 years, and 15 years, they are all fixed interest loans. The bank has $100 million in equity. It can borrow up to $900 million in demand deposits, 3 year certificates of deposit, or long term bonds. The spread between funds and loans of similar maturity is 2% at the moment they are made. (e.g. if demand deposits pay 1%, 6 months loans pay 3%)

A. Initially, the term structure of interest rates for funds is 1% for demand deposits, 2% for CDs, and 5% for 15 year bonds. What would a bank balance sheet look like it divided its funds equally across the three loan types and it wanted to match maturities? What is the banks net revenue in year 1. What is the return on equity?

Table 1: **Balance Sheet** Pasatech Bank 12/31/2024

Assets	**Amount (in $ million)**
6-Month Loans (Funded by Demand Deposits)	$333.3
3-Year Loans (Funded by Certificates of Deposit)	$333.3
15-Year Loans (Funded by Equity and Long-Term Bonds)	$333.3
Total Assets	$1,000
Liabilities	**Amount (in $ million)**
Demand Deposits	$300
Certificates of Deposit	$300
15 years Bonds	$300
Total Liabilities	$900
Equity	**Amount (in $ million)**
Equity	$100
Total Equity	$100
Total Liabilities and Equity	$1,000

Liabilities (Borrow at the following rates) costs the bank 24 which is the sum of:

(a) Demand Deposits 1 % : $300 \cdot 0.01 = 3$

(b) Certificate Deposits 2 %: $300 \cdot 0.02 = 6$

(c) Long Term Bonds 5 %: $300 \cdot 0.05 = 15$

Assets (Lend at the following rates) gives the bank 46.662 which is the sum of:

(a) 6 month Loans 3 %: $333.3 \cdot 0.03 = 9.99$

(b) 3 year Loans 4 %: $333.3 \cdot 0.04 = 13.332$

(c) 15 year Bonds 7 %: $333.3 \cdot 0.07 = 23.331$

Hence, the net revenue is $46.662 - 24 = 22.662$ (million dollars) and the return on equity is the net revenue over the equity, i.e. $22.662/100 \simeq 22.67\%$

B. Assuming inflows and outflows of demand deposits are equal in each year. What does a profit-maximizing balance sheet look like? What is the banks net revenue in year 1. What is the return on equity?

If the Bank wants to maximize profit they pick the liabilities with the lowest rate and the asset with the highest one. In this case the Balance Sheet would be the following:

Balance Sheet (profit-maximizing)

Pasatech Bank

12/31/2024

Assets	**Amount (in $ million)**
15-year Loans	$1,000
Total Assets	$1,000

Liabilities	**Amount (in $ million)**
Demand Deposits	$900
Equity	**Amount (in $ million)**
Total Equity	$100
Total Liabilities and Equity	$1,000

Liabilities (Borrow at the following rates) costs the bank 9 which is:

(a) Demand Deposits 1 % : $900 \cdot 0.01 = 9$

Assets (Lend at the following rates) gives the bank 70 which is:

(a) 15 year Loans 7 %: $1,000 \cdot 0.07 = 70$

Hence, the net revenue is $70 - 9 = 61$ (million dollars) and the return on equity is the net revenue over the equity, i.e. $61/100 = 61\%$

C. Suppose the bank followed the answer you gave for B. What happens to the bank's balance sheet, in year 2 if there is a sudden term structure inversion and the term structure of interest rates for funds is 6% for demand deposits 5.75% for CDs, and 5.5% for 15 year bonds?

If the Bank is stuck with the loans they made but they can adjust their liabilities they will now choose the one that with the lowest rates now. In this case the Balance Sheet would be the following:

Balance Sheet (profit-maximizing)
Pasatech Bank
12/31/2024

Assets	**Amount (in $ million)**
15-year Loans	$1,000
Total Assets	$1,000

Liabilities	**Amount (in $ million)**
15 year Bonds	$900
Equity	**Amount (in $ million)**
Total Equity	$100
Total Liabilities and Equity	$1,000

Liabilities (Borrow at the following rates) costs the bank 49.5 which is:

(a) 15 year Loan 5.5 % : $900 \cdot 0.01 = 49.5$

Assets (Lend at the following rates) gives the bank 70 which is:

(a) 15 year Loans 7 %: $1,000 \cdot 0.07 = 70$

Hence, the net revenue is $70 - 49.5 = 20.5$ (million dollars) and the return on equity is the net revenue over the equity, i.e. $20.5/100 = 20.5\%$

D. Returning to rates of question A and B. Suppose that every year, there is a 10% chance of a serious downturn, and bank customers will withdraw half the demand deposits. What is a balance sheet that avoids a bank run? And maximize profits

A bank run occurs when equity goes to zero so the most they can use of demand deposits is 200 million.

Balance Sheet
Pasatech Bank
12/31/2024

Assets	**Amount (in $ million)**
6-Month Loans (Funded by Demand Deposits)	$0
3-Year Loans (Funded by Certificates of Deposit)	$0
15-Year Loans (Funded by Equity and Long-Term Bonds)	$1,000
Total Assets	$1,000

Liabilities	**Amount (in $ million)**
Demand Deposits	$0
Certificates of Deposit	$900
15 years Bonds	$0
Total Liabilities	$900

Equity	**Amount (in $ million)**
Equity	$100
Total Equity	$100
Total Liabilities and Equity	$1,000

Liabilities (Borrow at the following rates) costs the bank 18 which is the sum of:

(a) Demand Deposits 1 % : $0 \cdot 0.01 = 0$

(b) Certificate Deposits 2 %: $900 \cdot 0.02 = 18$

(c) Long Term Bonds 5 %: $0 \cdot 0.05 = 0$

Assets (Lend at the following rates) gives the bank 70 which is the sum of:

(a) 6 month Loans 3 %: $0 \cdot 0.03 = 0$

(b) 3 year Loans 4 %: $0 \cdot 0.04 = 0$

(c) 15 year Loans 7 %: $1,000 \cdot 0.07 = 70$

Hence, the net revenue is $70 - 18 = 52$ (million dollars) and the return on equity is the net revenue over the equity, i.e. $52/100 = 52\%$

E. If the bank is risk neutral will it follow that plan? Does it matter how much equity the bank has?

If the bank takes the risky choice from B, 90 % the bank gets 61 and with 10 % the banks get 0, so the expected value of taking the risk is

$$EV = 0.9 \cdot 61 - 0.1 \cdot 0 = 54.9$$

Since $54.9 > 52$, the risk neutral bank takes the risky option.

- Equity does not change this choice under risk neutrality: increasing equity scales both gains and losses but does not change which strategy has higher expected value.
- Equity does matter for survival: more equity lets the bank absorb deposit withdrawals and avoid failure, but a risk-neutral bank treats survival only as a constraint, not as an objective.

3 Mean and Variance

A. In the CSV file that is also posted there is a price series for 3 stocks. Using excel or a statistical package of your choice, compute the mean return. Remember to annualize it so you can compare over time for each of the stocks for three different units of time: one year, over the last three years, and the last ten years.

Calculate the daily return as

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

```
data <- data[-c(1259:1261),]

dr1 <- rep(NA, nrow(data))
dr2 <- rep(NA, nrow(data))
dr3 <- rep(NA, nrow(data))

for(i in 2:nrow(data)){
  dr1[i] = (data$S1[i] - data$S1[i-1])/data$S1[i-1]
  dr2[i] = (data$S2[i] - data$S2[i-1])/data$S2[i-1]
  dr3[i] = (data$S3[i] - data$S3[i-1])/data$S3[i-1]
}

daily_rates <- data.frame(cbind(data$Date, dr1, dr2, dr3))

colnames(daily_rates) <- c('Date', 'dr1', 'dr2', 'dr3')

daily_rates[1:5,]
```

```
##        Date                   dr1                  dr2                  dr3
## 1 1/10/2018                  <NA>                 <NA>                 <NA>
## 2 1/11/2018 -0.000372614999999965 -0.00492176499999997 0.00602772800000007
## 3 1/12/2018   -0.0447356281660891   0.0169267193347867  0.0185141467591836
## 4 1/16/2018   -0.0054635448362878  0.00270211830482208  0.0171185883444519
## 5 1/17/2018  -0.00442845929122344   0.0133125042439877 -0.0207055356811494
```

Then take the average over the given time period.

(a) **Year to date:**

```
year_index <- which(daily_rates$Date == '1/10/2022')

year_dr <- data.frame(lapply(daily_rates[year_index:1258, 2:4], as.numeric))

avg_returns <- colMeans(year_dr, na.rm = T)

avg_returns
```

```
##           dr1           dr2           dr3
## -0.0028770829  0.0007834205  0.0006598247
```

Now annualise it using the formula

$$(1 + r)^{365} - 1$$

```r
(1+avg_returns)^365 - 1
```

```
##         dr1         dr2         dr3
## -0.6506388  0.3308749  0.2722113
```

(b) **Last 3 years**

```r
year3_index <- which(daily_rates$Date == '1/9/2020')

year3_dr <- data.frame(lapply(daily_rates[year3_index:1258, 2:4], as.numeric))

avg_returns3 <- colMeans(year3_dr, na.rm = T)

avg_returns3
```

```
##           dr1           dr2           dr3
## -0.0001840629  0.0003207505  0.0009442674
```

Now annualise:

```r
(1+avg_returns3)^365 - 1
```

```
##         dr1         dr2         dr3
## -0.06498167  0.12418146  0.41127699
```

(c) **Last 10 years:**

```r
year10_dr <- data.frame(lapply(daily_rates[, 2:4], as.numeric))

avg_returns10 <- colMeans(year10_dr, na.rm = T)

avg_returns10
```

```
##           dr1           dr2           dr3
## 8.768556e-05 4.427337e-04 5.483768e-04
```

Now annualise:

```r
(1+avg_returns10)^365 - 1
```

```
##         dr1         dr2         dr3
## 0.03252146 0.17534536 0.22152818
```

B. Compute the variance of these stocks for the same three units of time.

(a) **Year to date:**

```
year_dr %>%
  summarise(v1 = var(dr1, na.rm = T), v2 = var(dr2, na.rm = T), v3 = var(dr3, na.rm = T))
```

```
##            v1           v2           v3
## 1 0.001641876 0.0001931756 0.0004247779
```

(b) **Last 3 years**

```
year3_dr %>%
  summarise(v1 = var(dr1, na.rm = T), v2 = var(dr2, na.rm = T), v3 = var(dr3, na.rm = T))
```

```
##             v1           v2           v3
## 1 0.0009433312 0.0003114824 0.0004922444
```

(c) **Last 10 years:**

```
year10_dr %>%
  summarise(v1 = var(dr1, na.rm = T), v2 = var(dr2, na.rm = T), v3 = var(dr3, na.rm = T))
```

```
##             v1           v2           v3
## 1 0.0007436644 0.0002659788 0.0004503822
```

C. Which units of time would you choose to measure mean and variance when making an investment decision (among these three stocks) if you plan to cash out in three months?

Cashing out in three months suggests looking at the most recent trends. This leans towards the year to date values.

D. Would your answer change if your horizon was six years? What assumptions underlie your answer?

The further in the future you are looking to cash out, the longer the trend you are interested in. Therefore, this lends itself more towards the 10 year trend. Obviously you hope that there have been no significant shocks or innovations that have affected the longer trends.

4 Riskless Bonds

We will use the following formula: $V = \sum_{t=1}^{T} \frac{C}{(1+r)^t} + \frac{1}{(1+r)^T}$ for the price of the bond as a proportion of its face value.

A. The yield on the 1-year zero-coupon T-Bill was 0.4% on December 1, 2020; 2.16% on June 1, 2022; and 4.44% on June 1, 2023. What was its price in each period?

(a) **Dec 2020:** $\frac{1}{(1+0.004)^1} = 0.996$

(b) **June 2022:** $\frac{1}{(1+0.0216)^1} = 0.978$

(c) **June 2023:** $\frac{1}{(1+0.0444)^1} = 0.957$

B. The yield on the 5-year T-Bill was 0.42% on December 1, 2020; 2.94% on June 1, 2022; and 3.7% on June 1, 2023. What was its price if it had a zero coupon? What would be its price if it had a 1% coupon?

Case 1: $C = 0$

(a) **Dec 2020:** $\frac{1}{(1+0.0042)^5} = 0.978$

(b) **June 2022:** $\frac{1}{(1+0.0294)^5} = 0.866$

(c) **June 2023:** $\frac{1}{(1+0.037)^5} = 0.835$

Case 2: $C = 0.01$

(a) **Dec 2020:** $\sum_{t=1}^{5} \frac{0.01}{(1+0042)^t} + \frac{1}{(1+0.0042)^5} = 1.03$

(b) **June 2022:** $\sum_{t=1}^{5} \frac{0.01}{(1+0.0294)^t} + \frac{1}{(1+0.0294)^5} = 0.91$

(c) **June 2023:** $\sum_{t=1}^{5} \frac{0.01}{(1+0.037)^t} + \frac{1}{(1+0.037)^5} = 0.879$

C. For December 1, 2020, and June 1, 2023, compute the expected constant interest rate for years 2 to 5.

Denote the constant interest rate for years 2 to 5 as $f_{1,5}$. The expectations hypothesis implies : $(1 + r_1)(1 + f_{1,5})^4 = (1 + r_5)^5$ which implies

$$f_{1,5} = \left(\frac{(1 + r_5)^5}{1 + r_1}\right)^{1/4} - 1$$

(a) **Dec 2020:** $f_{1,5} = \left(\frac{(1+0.0042)^5}{1+0.004}\right)^{1/4} - 1 = 0.00425$

(b) **June 2023:** $f_{1,5} = \left(\frac{(1+0.037)^5}{1+0.0444}\right)^{1/4} - 1 = 0.035$

D. On July 31, 2024, the yield on the 30-year T-Bill was 4.35%, down from 4.9% on June 8. If those T-Bills have a 4% coupon (paid annually), by how much did a $10,000 investment change in value?

(a) **July 31 2024:** $\sum_{t=1}^{30} \frac{0.04}{(1+0435)^t} + \frac{1}{(1+0.0435)^{30}} = 0.942$

(b) **June 8 2024:** $\sum_{t=1}^{30} \frac{0.04}{(1+0.049)^t} + \frac{1}{(1+0.049)^{30}} = 0.86$

$$10,000 \cdot 0.942 - 10,0000 \cdot 0.86 = 820$$

5 Risk Aversion

An investor who has \$10 is faced with three possible assets, each priced at \$1.

- **Asset 1:** If there is above-average rainfall in the next six months, the first asset pays \$2. If there is below-average rainfall, it pays \$0.5.
- **Asset 2:** Pays \$1.5 if there is above-average rainfall and \$0.8 if rainfall is below average.
- **Asset 3:** Pays \$0.75 if rainfall is above average and \$1.5 if it is below average.

1. Suppose the distribution of rainfall is symmetric about the mean. What is the expected value of the assets and their variance?

- **Asset 1:**

$$\begin{aligned} \mathrm{E} &= (0.5)(0.75) + (0.5)(1.5) = 1.125 \\ \mathrm{E}X^2 &= (0.5)(2^2 + 0.5^2) = 2.125 \\ \mathrm{Var} &= 2.125 - (1.125)^2 = 0.5626 \end{aligned}$$

- **Asset 2:**

$$\begin{aligned} \mathrm{E} &= (0.5)(1.5) + (0.5)(0.8) = 1.15 \\ \mathrm{E}X^2 &= (0.5)(1.5^2 + 0.8^2) = 1.445 \\ \mathrm{Var} &= 1.445 - (1.15)^2 = 0.1225 \end{aligned}$$

- **Asset 3:**

$$\begin{aligned} \mathrm{E} &= (0.5)(0.75) + (0.5)(1.5) = 1.125 \\ \mathrm{E}X^2 &= (0.5)(0.75^2 + 1.5^2) = 1.40625 \\ \mathrm{Var} &= 1.40625 - (1.125)^2 = 0.140625 \end{aligned}$$

2. Suppose the mean rainfall is much higher than the median. What happens to the expected value of the assets?

   When the mean rainfall is much higher than the median, the probability of above-average rainfall decreases. Let $p < 0.5$ represent the probability of above-average rainfall and $1 - p$ represent the probability of below-average rainfall.

   - **Asset 1:** $\mathrm{E} = p(2) + (1-p)(0.5) = 0.5 + 1.5p$
   - **Asset 2:** $\mathrm{E} = p(1.5) + (1-p)(0.8) = 0.8 + 0.7p$
   - **Asset 3:** $\mathrm{E} = p(0.75) + (1-p)(1.5) = 1.5 - 0.75p$

   As p increases, Asset 1 gains the most in expected value.

3. Returning to the symmetric case, would you expect investors who only have access to the first two assets to split their investments between them? If not, why?

   - Asset 1 has a higher expected value (\$1.25 vs. \$1.15) but also higher variance (0.5625 vs. 0.0625).
   - Risk-averse investors may prioritize Asset 2 for its lower variance.
   - Risk-neutral investors will favor Asset 1 because it maximizes Expected Value.

4. Under what circumstances would someone choose Asset 2 over Asset 1?

   They are risk-averse enough that they prioritize lower variance over higher returns.

5. If Asset 3 is available, what mix of assets minimizes variance?

   The optimal mix of assets can be determined by minimizing portfolio variance:

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + w_3^2\sigma_3^2$$
$$\text{subject to } w_1 + w_2 + w_3 = 1 \text{ and } 0 \leq w_1, w_2, w_3 \leq 1.$$

The optimal weights are:

$$w_1 = 0.071 \quad (7.1\%)$$
$$w_2 = 0.643 \quad (64.3\%)$$
$$w_3 = 0.285 \quad (28.5\%)$$

The minimized portfolio variance is $\sigma_p^2 = 0.0402$.

6 Debt and Equity

The Rose and Thal real estate company has been building malls for the past twenty years. It has \$3 billion in outstanding bonds that pay 3%. \$500 million matures each year for the next six years. When the bonds mature, the company refinances at the prevailing interest rate. The company's profit before paying interest is \$200 million.

A. What would the value of the equity be if this profit was earned for 30 years, assuming a discount rate of 0.95? Assume interest rates remain stable at 3%.

   Each year the company makes a profit of \$200m and pays 3% on the outstanding \$3 billion bonds. Therefore the yearly profit after interest is $200 - 0.03 \cdot 3000 = 110$ millions. Discounting this for 30 years, we have:

$$\sum_{t=1}^{30} 0.95^t (110) \simeq 1.73 \text{ billion}$$

B. Interest rates have jumped, and as the bonds mature, the company must refinance at 6% (each time a packet of bonds mature). What is the value of its equity using the same discount rate of 0.95?

   Interest Payments

   The firm has \$3 billion in outstanding debt. Each year for six years, \$0.5 billion matures and is refinanced at an interest rate of 6%, while the remaining debt continues to pay 3%. The firm's profit before interest is \$200 million per year. The discount factor is $d = 0.95$.

In year $t = 1, \ldots, 6$:

$$\text{Debt at } 6\% = 0.5t \text{ (billions)},$$
$$\text{Debt at } 3\% = 3 - 0.5t \text{ (billions)}.$$

Thus, interest payments in year t are:

$$\begin{aligned} I_t &= 0.06(0.5t) + 0.03(3 - 0.5t) \\ &= 0.09 + 0.015t \quad \text{(billions)}. \end{aligned}$$

In millions: $I_t = 90 + 15t$.

Profit to Equity

For $t = 1, \ldots, 6$:

$$\pi_t = 200 - (90 + 15t) = 110 - 15t.$$

After year 6, all debt is refinanced at 6%:

$$I = 0.06 \times 3{,}000 = 180,$$

so for $t = 7, \ldots, 30$:

$$\pi_t = 200 - 180 = 20.$$

Present Value of Equity

The value of equity is the present value of profits to equity:

$$V_E = \sum_{t=1}^{6} 0.95^t (110 - 15t) + \sum_{t=7}^{30} 0.95^t (20).$$

The first term evaluates to approximately:

$$\sum_{t=1}^{6} 0.95^t (110 - 15t) \approx 300.7.$$

The second term is a discounted annuity:

$$\sum_{t=7}^{30} 0.95^t (20) = 20 \cdot 0.95^7 \cdot \frac{1 - 0.95^{24}}{0.05} \approx 197.8.$$

Hence, the total value is around 497.8 million dollars

C. Thal, who went to Caltech, argues that the discount rate should be 0.943. Why? (*Hint: What is the opportunity cost of capital?*)

Thal is right because the opportunity cost of capital is 6%. Therefore, equity cash flows should be discounted at 6%, which implies a discount factor of: $1/1.06 = 0.943$.

D. Suppose Rose, the chief risk officer, is considering the possibility of interest rates rising to 8%. What would the equity value be? Who owns the firm? Each year for six years, \$0.5 billion of debt is refinanced at 8%, while the remaining debt continues to pay 3%. Profit before interest is \$200 million.

Interest Payments In year $t = 1, \ldots, 6$:

$$I_t = 0.08(0.5t) + 0.03(3 - 0.5t) = 0.09 + 0.025t \quad \text{(billions)}.$$

In millions:

$$I_t = 90 + 25t.$$

Profit to Equity

$$\pi_t = 200 - (90 + 25t) = 110 - 25t.$$

Equity cash flows become negative in year 5, implying default. Thus equity receives cash flows only for $t = 1, \ldots, 4$:

$$(\pi_1, \pi_2, \pi_3, \pi_4) = (85, 60, 35, 10).$$

Equity Value Using the opportunity cost of capital $r = 8\%$ (discount factor $d = 1/1.08$):

$$V_E = \sum_{t=1}^{4} \left(\frac{1}{1.08}\right)^t (110 - 25t) \approx 0.17 \text{ billion}.$$

Ownership Once the firm cannot service its debt, equity is wiped out. The firm is effectively owned by its bondholders.

E. Suppose Rose and Thal start their company and need to raise \$3 billion. In 20 years, they can sell the company for \$5 billion. In the meantime, they will earn \$200 million a year. They discount the future at 5% per year. If they borrow \$3 billion at an interest

rate of 5% for 20 years, what is the net present value of the firm? They can also raise capital by issuing shares to investors who expect a 5.5 % on equity. What fraction of the firm must they give to the equity investors. What do they get to keep? What is Rose and Thal's share worth? Do they prefer debt or equity? Why?

Rose and Thal need to raise \$3 billion to start a company. The firm earns \$200 million per year for 20 years and can be sold for \$5 billion at $t = 20$. Future cash flows are discounted at 5%.

Net Present Value of the Firm

The present value of operating cash flows and the terminal value is:

$$PV = \sum_{t=1}^{20} \frac{200}{(1.05)^t} + \frac{5000}{(1.05)^{20}} = 200 \cdot \frac{1-(1.05)^{-20}}{0.05} + \frac{5000}{(1.05)^{20}} \approx 4376.9,$$

where values are in millions.
Subtracting the initial investment:

$$NPV = 4376.9 - 3000 \approx 1376.9.$$

Thus,

$$NPV \approx \$1.38 \text{ billion}$$

Equity Financing

Suppose instead the firm raises \$3 billion by issuing equity. Equity investors require a return of 5.5%. The value of the firm using this discount rate is:

$$V_0 = 200 \cdot \frac{1-(1.055)^{-20}}{0.055} + \frac{5000}{(1.055)^{20}} \approx 4103.7.$$

Let f denote the fraction of the firm sold to investors:

$$3000 = f \cdot 4103.7 \quad \Rightarrow \quad f \approx 0.731.$$

Thus, equity investors receive about 73.1% of the firm, and Rose and Thal keep:

$$1 - f \approx 26.9\%.$$

The value of Rose and Thal's retained stake is:

$$V_{\text{R\&T}} = 4103.7 - 3000 \approx 1103.7.$$

Financing Choice

Under debt financing at 5%, Rose and Thal capture the full project surplus:

$$\text{Value to founders} \approx \$1.38 \text{ billion}.$$

Under equity financing at 5.5%, their stake is worth only:

$$\text{Value to founders} \approx \$1.10 \text{ billion}.$$

Conclusion Rose and Thal prefer debt financing because it has a lower required return than equity. Debt is cheaper capital, allowing them to retain a larger share of the firm's value, although it exposes them to greater risk.